DOCKHUB · REGULATORY REGISTER

DockHub regulatory and standards register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-31	Amish Chadha	First register at the kit 1.9.1 upgrade, researched for the current design (option A) with New York City as the first candidate city (DKH-DEC-001); sources cited with the date checked

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A street-side station that stores and charges lithium-ion packs unattended is regulated mostly through fire, electrical and public-space rules, not through a single product certificate. New York City, the first candidate city, shows the pattern: its Department of Transportation announced in November 2025 a citywide public battery-swapping network of 25 flagship sites, to be placed only in cabinets "properly certified to applicable UL standards" and cleared by the Fire Department (FDNY), with fire suppression, real-time battery health monitoring and automatic alerts. The FDNY guide for battery charging cabinets then sets the certification route (UL 1487 fire testing, a hazard mitigation analysis, a Certificate of Approval and, outdoors, a site survey and a DOT Revocable Consent on public property). The same guide does not mention swapping stations by name, so whether a DockHub cabinet is treated as a charging cabinet or needs its own approval is **to assess** with FDNY. DockHub's first pilot is on private property (DKH-DEC-001), which avoids the street consent but not the fire review. Not legal advice; every row marked "to assess" needs a qualified reviewer, and rules differ by jurisdiction. Nothing here needs certification before TRL 4.

Table 1. Regulatory and standards register.

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
G1	FDNY Certificate of Approval for battery charging cabinets (TM-2 application, UL 1487 test report, hazard mitigation analysis by a registered design professional); fees listed by FDNY: $625 for the application, $420 for an outdoor site application	New York City	Approval of an enclosure that charges battery packs	The cabinet charges packs in an enclosed cabinet, which the Fire Code allows only with FDNY approval; whether a swap cabinet follows the same route is not stated	Design lead with a registered design professional	To assess with FDNY
G2	UL 1487, battery containment enclosures	United States (cited by FDNY)	Fire containment of a battery enclosure, tested by a national laboratory	The bay compartments and cabinet are a battery containment design; no test planned before TRL 4	Design lead	Certification needed later
G3	UL 4900, micromobility charging equipment; UL 2849, UL 2272 and UL 2271 for e-bike systems and batteries	United States (cited by FDNY)	Listing of charging equipment and the packs it charges	Chargers and the SwapCell pack should be listed; the pack's listing is the SwapCell project's	SwapCell project; design lead	To assess
G4	NYC DOT public e-bike battery swapping network (announced 2025-11-18): certified cabinets, fire suppression, real-time battery health monitoring, alerts to the system provider; membership access, prices not yet stated	New York City	Rules for cabinets placed on the city's sidewalks	DockHub's design has detection, suppression, health reporting and alerts; certification is the open item	Co-design partner	To assess
G5	NYC DOT Revocable Consent for equipment on public property; FDNY Site Survey Inspection, Letter of No Objection and DOB electrical permit (ED16A) for outdoor installations	New York City	Siting and electrical work	Needed for a public sidewalk site, not for the first private pilot's circuit review beyond the electrical permit	Co-design partner	To assess; first pilot on private property
G6	NFPA 855, installation of stationary energy storage systems; UL	United States (named by	Spacing, protection and a propagation test	The aerosol versus vermiculite trial (DKH-DEC-001) aims at a UL	Design lead	To assess

#	Standard or regulation	Jurisdiction	What it covers	Applicability to this design	Owner (role)	Status
	9540A, thermal runaway propagation test method	number; texts not consulted)	for battery storage	9540A style unit-level test before an accredited lab; applicability of NFPA 855 to about 2 kWh of packs is unknown		
G7	NFPA 70 (National Electrical Code) and local electrical rules; the grid input is a single circuit with an RCBO	United States (named by number; text not consulted)	Wiring, protection, outdoor equipment	One 120 V 15 A or 230 V 10 A circuit; installation by a licensed electrical contractor	Licensed electrician	To assess
G8	CPSC proposed safety standard for lithium-ion batteries used in micromobility products (Federal Register, 2026-06-24, document 2026-12749)	United States	Safety of micromobility batteries and their electrical systems	Bears on the packs; DockHub does not make packs	SwapCell project	To assess
G9	ANSI/NEMA 250 enclosure type designations and IEC 60529 IP codes	United States; international (named; texts not consulted)	Weather and dust protection of enclosures	R9 asks for IP54; FDNY refers to NEMA 250 weatherization for cabinets	Design lead	To assess; test at TRL 4
G10	ADA Standards, operable parts: reach range 15 to 48 in above the floor	United States	Reachable controls	All operable parts at 0.73 to 1.10 m (PR-011)	Design lead	Met by design
G11	Radio and mains-EMC approval of the modem and chargers (FCC or CE marks, as the bought parts carry)	Country of deployment	Radio equipment	Bought modem and chargers must already be marked; DockHub adds no radio design	Design lead	To assess when parts are chosen
G12	Data protection for the rider token and swap log	Country of deployment (for example GDPR in the EU)	Personal data	No cameras or microphones; logs hold an opaque token (PR-012); whether a token is personal data is to assess	Co-design partner	To assess
G13	Manual handling and lone work for the technician and the pack lift	Country of deployment	Lifting and site safety	A 2.85 kg pack; the cabinet is installed with a lifting aid (the 1.1 t pad needs a crane or ready-mix delivery)	Design lead	To assess

Status: to assess, applies, does not apply, met by design, certification needed later.

Sources

- NYC Department of Transportation, press release "NYC DOT takes first steps to install public e-bike battery swapping network citywide", 2025-11-18, https://www.nyc.gov/html/dot/html/pr2025/install-public-e-bike-battery-swapping-network-citywide.shtml, accessed 2026-10-31 (G4, G5).
- FDNY, *E-Micromobility Approval Guide* (battery charging cabinets, UL 1487, UL 4900, COA and outdoor site survey, fees), https://www.nyc.gov/assets/fdny/downloads/pdf/business/e-micromobility-approval-guide.pdf, accessed 2026-10-31 (G1, G2, G3, G5, G9).
- CleanTechnica, "Battery Swapping Expands In New York City", 2026-02-28, https://cleantechnica.com/2026/02/28/battery-swapping-expands-in-new-york-city/, accessed 2026-10-31 (pilot and citywide plan; no standard named).
- US Access Board, *Guide to the ADA Standards, Chapter 3: Operable Parts*, https://www.access-board.gov/ada/guides/chapter-3-operable-parts/, accessed 2026-10-31 (G10).
- US Consumer Product Safety Commission via the Federal Register, 2026-06-24, https://www.federalregister.gov/documents/2026/06/24/2026-12749/safety-standard-for-lithium-ion-batteries-used-in-micromobility-products-and-electrical-systems-of, as cited in DKH-PRB-001 (G8); the document was not re-read for this register.
- Standards named by number only (G6, G7, G9 NFPA, IEC, UL 9540A, UL 2849, UL 2272, UL 2271): texts not consulted; each applicability is to be confirmed by the role named.